

Chapter Two

Theory of Demand and Supply

Introduction

Having learnt about the concept and meaning of economics as a subject and its nature, scope, different systems and various other fundamentals in the previous chapter, we now resort to a very important issue in economics. This is the issue of how free markets operate. In this chapter we will forward our exploration and understanding of the vast field of economics by focusing on two very powerful tools, namely, theory of demand and theory of supply.

The purpose of this chapter is to explain what demand and supply are and show how they determine equilibrium price and quantity. We will also show how the concepts of demand and supply reveal consumers' and producers' sensitivity to price change.

Chapter objectives

After covering this chapter, you will be able to:

- understand the concept of demand and the factors affecting it;
- explain the supply side of a market and the determinants of supply;
- understand how the market reaches equilibrium condition, and the possible factors that could cause a change in equilibrium and
- explain the elasticity of demand and supply

2.1 Theory of demand

1. *Are demand and want similar? Why?*
2. *Why can't we purchase all that we need or we desire to have?*
3. *Can we say that, with a decrease in the price of a commodity, a consumer normally buys more of it? Why?*

4. *Explain why demand curves always slope downwards from left to right. Are there any exceptions to this?*

Demand is one of the forces determining prices. The theory of demand is related to the economic activities of consumers-consumption. Hence, the purpose of the theory of demand is to determine the various factors that affect demand.

In our day-to-day life we use the word ‘demand’ in a loose sense to mean a desire of a person to purchase a commodity or service. But in economics it has a specific meaning, which is different from what we use it in our day to day activities.

Demand implies more than a mere desire to purchase a commodity. It states that the consumer must be willing and able to purchase the commodity, which he/she desires. His/her desire should be backed by his/her purchasing power. A poor person is willing to buy a car; it has no significance, since he/she has no ability to pay for it. On the other hand, if his/her desire to buy the car is backed by the purchasing power then this constitutes demand. Demand, thus, means the desire of the consumer for a commodity backed by purchasing power. These two factors are essential. If a consumer is willing to buy but is not able to pay, his/her desire will not become demand. Similarly, if the consumer has the ability to pay but is not willing to pay, his/her desire will not be called demand.

More specifically, **demand** refers to various quantities of a commodity or service that a consumer would purchase at a given time in a market at various prices, given other things unchanged (*ceteris paribus*). The quantity demanded of a particular commodity depends on the price of that commodity.

Law of demand: This is the principle of demand, which states that , price of a commodity and its quantity demanded are inversely related i.e., as price of a commodity increases (decreases) quantity demanded for that commodity decreases (increases), *ceteris paribus*.

2.1.1 Demand schedule (table), demand curve and demand function

The relationship that exists between price and the amount of a commodity purchased can be represented by a table (schedule) or a curve or an equation.

Demand schedule can be constructed for any commodity if the list of prices and quantities purchased at those prices are known. An individual demand schedule is a list of the various quantities of a commodity, which an individual consumer purchases at various levels of prices in the market. A demand schedule states the relationship between price and quantity demanded in a table form.

Table 2.1 Individual household demand for orange per week

Combinations	A	B	C	D	E
Price per kg	5	4	3	2	1
Quantity demand/week	5	7	9	11	13

Demand curve is a graphical representation of the relationship between different quantities of a commodity demanded by an individual at different prices per time period.

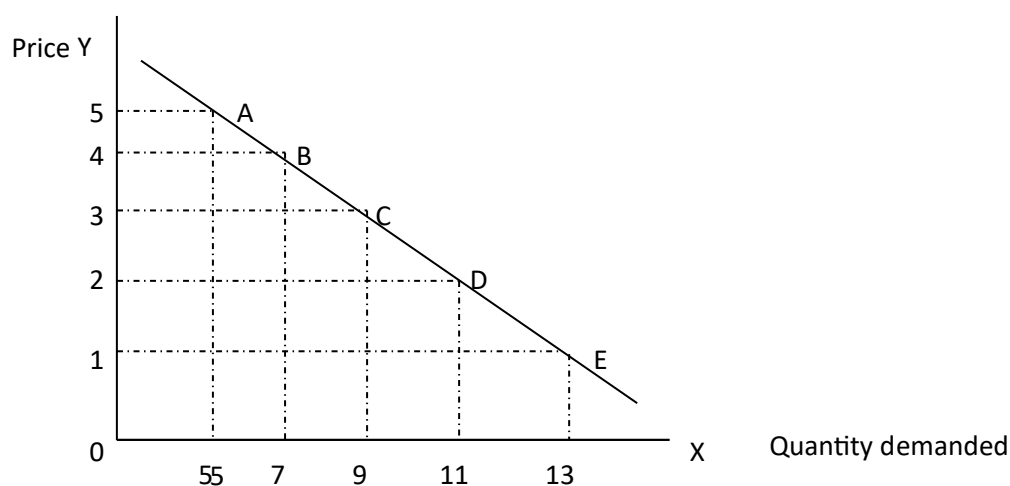


Figure 2.1: Individual demand curve

In the above diagram prices of oranges are given on OY axis and quantity demanded on OX axis. For example, when the price per kilogram is birr 1 the quantity demanded is 13 kilograms. From the above figure you may notice that as the price declines quantity demanded increases and vice-versa.

Demand function is a mathematical relationship between price and quantity demanded, all other things remaining the same. A typical demand function is given by: $Q_d=f(P)$ where Q_d is quantity demanded and P is price of the commodity, in our case price of orange. Example: Let the demand function be $Q = a + bP$

$b = \frac{\Delta Q}{\Delta P}$ (e.g. moving from point A to B on figure 2.1 above) P

$b = \frac{7-5}{4-5} = -2$, where b is the slope of the demand curve

$Q = a - 2P$, to find a , substitute price either at point A or B. $7 =$

$a - 2(4)$, $a = 15$

Therefore, $Q = 15 - 2P$ is the demand function for orange in the above numerical example.

Market Demand: The market demand schedule, curve or function is derived by horizontally adding the quantity demanded for the product by all buyers at each price.

Table 2.2: Individual and market demand for a commodity

Price	Individual demand			Market demand
	Consumer-1	Consumer-2	Consumer-3	
8	0	0	0	0
5	3	5	1	9
3	5	7	2	14
0	7	9	4	20

The following graph depicts market demand curve at price equal to 3

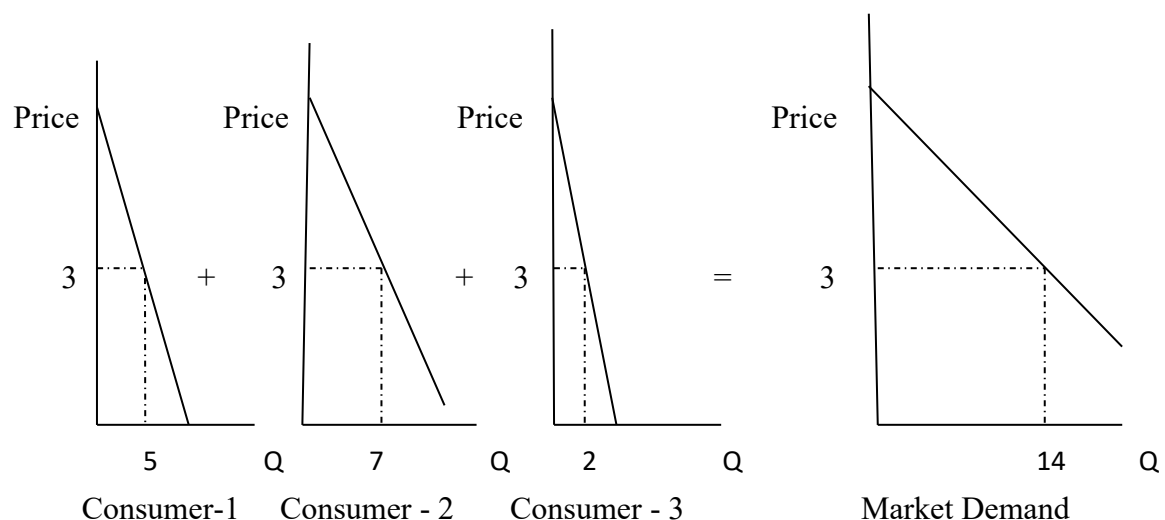


Figure 2.2: Individual and Market demand curve

Numerical Example: Suppose the individual demand function of a product is given by: $P=10 - Q/2$ and there are about 100 identical buyers in the market. Then the market demand function is given by:

$$P= 10 - Q/2 \leftrightarrow Q/2=10-P \leftrightarrow Q= 20 - 2P \text{ and } Q_m = (20 - 2P) 100 = \underline{2000-200P}$$

2.1.2 Determinants of demand

The demand for a product is influenced by many factors. Some of these factors are:

- I. Price of the product
- II. Taste or preference of consumers
- III. Income of the consumers
- IV. Price of related goods
- V. Consumers expectation of income and price
- VI. Number of buyers in the market

When we state the law of demand, we kept all the factors to remain constant except the price of the good. A change in any of the above listed factors except the price of the good will change the demand, while a change in the price, other factors remain constant will bring change in quantity demanded. A change in demand will shift the demand curve from its original location. For this reason those factors listed above other than price are called demand shifters. A change in own price is only a movement along the same demand curve.

Changes in demand: a change in any determinant of demand—except for the good's price—causes the demand curve to shift. We call this a change in demand. If buyers choose to purchase more at any price, the demand curve shifts rightward—an increase in demand. If buyers choose to purchase less at any price, the demand curve shifts leftward—a decrease in demand.

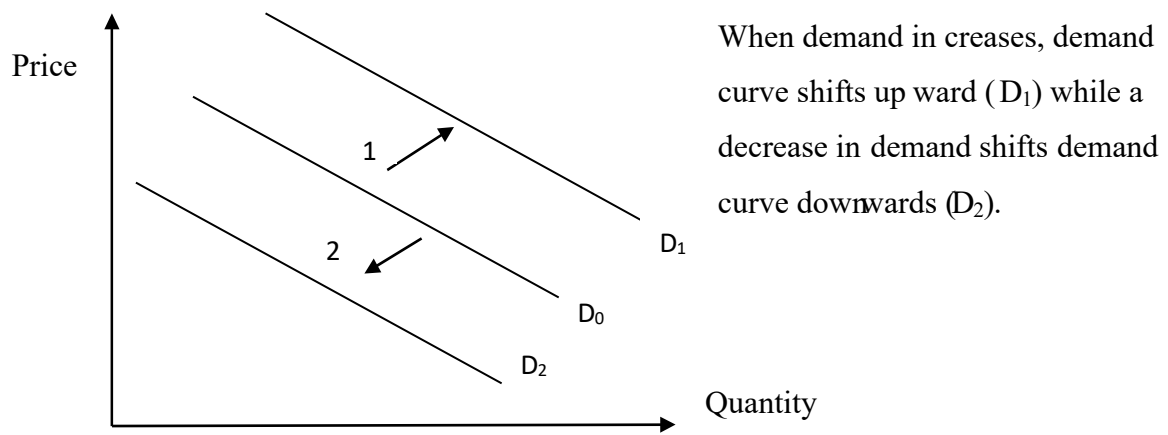


Figure 2.3: Shift in demand curve Now

let us examine how each factor affect demand.

I. Taste or preference

When the taste of a consumer changes in favour of a good, her/his demand will increase and the opposite is true.

II. Income of the consumer

Goods are classified into two categories depending on how a change in income affects their demand. These are normal goods and inferior goods. **Normal Goods** are goods whose demand increases as income increase, while **inferior goods** are those whose demand is inversely related with income. In general, inferior goods are poor quality goods with relatively lower price and buyers of such goods are expected to shift to better quality goods as their income increases. However, the classification of goods into normal and inferior is subjective and it is usually dependent on the socio-economic development of the nation.

III. Price of related goods

Two goods are said to be related if a change in the price of one good affects the demand for another good.

There are two types of related goods. These are substitute and complimentary goods. **Substitute goods** are goods which satisfy the same desire of the consumer. For example, tea and coffee or Pepsi and Coca-Cola are substitute goods. If two goods are substitutes, then price of one and the demand for the other are directly related. **Complimentary goods**, on the other hand, are those goods which are jointly consumed. For example, car and fuel or tea and sugar are

considered as compliments. If two goods are complements, then price of one and the demand for the other are inversely related.

IV. Consumer expectation of income and price

Higher price expectation will increase demand while a lower future price expectation will decrease the demand for the good.

V. Number of buyer in the market

Since market demand is the horizontal sum of individual demand, an increase in the number of buyers will increase demand while a decrease in the number of buyers will decrease demand.

2.1.3 Elasticity of demand

1. *List some goods/commodities you think that increase in their prices will not significantly decrease their quantity demanded.*
2. *Can you list some products for which increase in their prices will significantly decrease/increase their quantity demanded?*

In economics, the concept of elasticity is very crucial and is used to analyze the quantitative relationship between price and quantity purchased or sold. Elasticity is a measure of responsiveness of a dependent variable to changes in an independent variable. Accordingly, we have the concepts of elasticity of demand and elasticity of supply.

Elasticity of demand refers to the degree of responsiveness of quantity demanded of a good to a change in its price, or change in income, or change in prices of related goods. Commonly, there are three kinds of demand elasticity: price elasticity, income elasticity, and cross elasticity.

A. Price Elasticity of Demand

Price elasticity of demand means degree of responsiveness of demand to change in price. It indicates how consumers react to changes in price. The greater the reaction the greater will be the elasticity, and the lesser the reaction, the smaller will be the elasticity. Price elasticity of demand is a measure of how much the quantity demanded of a good responds to a change in

the price of that good, computed as the percentage change in quantity demanded divided by the percentage change in price.

Demand for commodities like clothes, fruit etc. changes when there is even a small change in their price, whereas demand for commodities which are basic necessities of life, like salt, food grains etc., may not change even if price changes, or it may change, but not in proportion to the change in price.

Price elasticity demand can be measured in two ways. These are point and arc elasticity.

a. Point Price Elasticity of Demand

This is calculated to find elasticity at a given point. The price elasticity of demand can be determined by the following formula.

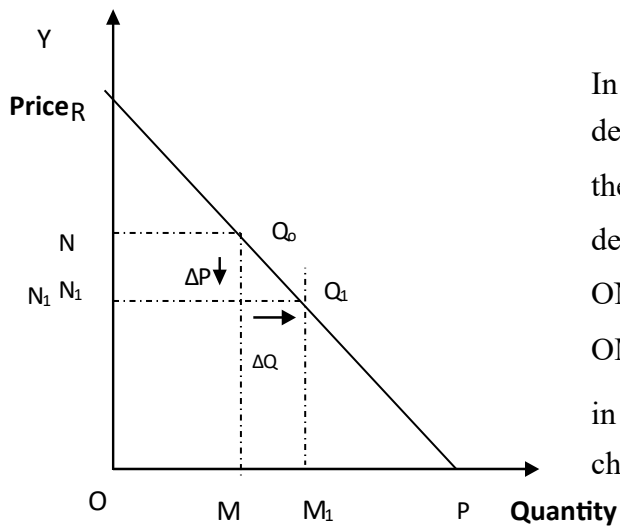
$$Ed = \frac{\text{Percentage change in quantity demanded}}{\text{percentage change in price}}$$

or

$$Ed = \frac{\text{Change in quantity demanded}}{\text{Change in price}} \times \frac{\text{Original price}}{\text{Original demanded}}$$

$$Ed = \left(\frac{Q_2 - Q_1}{p_2 - p_1} \right) \left(\frac{P_0}{Q_0} \right)$$

In this method, we take a straight-line demand curve joining the two axes, and measure the elasticity between two points Q_0 and Q_1 which are assumed to be intimately close to each other.



In the diagram RP' is the straight line demand curve, which connects both axes. In the beginning at the price ON the quantity demanded is OM . Then the price changes to ON_1 and the new quantity demanded will be OM_1 . The symbol ΔP represents the change in price while the symbol ΔQ shows the change in quantity demanded.

Figure 2.4 Point elasticity of demand

On a straight-line demand curve we can make use of this formula to find out the price elasticity at any particular point. We can find out numerical elasticities also on different points of the demand curve with the help of the above formula. It should be remembered that the point elasticity of demand on a straight line is different at every point.

b. Arc price elasticity of demand

The main drawback of the point elasticity method is that it is applicable only when we have information about even the slight changes in the price and the quantity demanded of the commodity. But in practice, we do not acquire such information about minute changes. We may possess demand schedules in which there are big gaps in price as well as the quantity demanded. In such cases, there is an alternative method known as arc method of elasticity measurement.

In arc price elasticity of demand, the midpoints of the old and the new values of both price and quantity demanded are used. It measures a portion or a segment of the demand curve between the two points. An arc is a portion of a curve line, hence, a portion or segment of a demand curve.

The formula for measuring arc elasticity is given below.

$$E_d = \frac{\text{Change in quantity demanded}}{\text{Change in price}} \times \frac{\text{Original price plus new price}}{\text{Original quantity plus new quantity demanded}}$$

Symbolically, the formula may be expressed thus:

$$E_d = \left(\frac{Q_1 - Q_0}{P_1 - P_0} \right) \times \frac{(P_0 + P_1)}{(Q_0 + Q_1)}$$

Here, Q_0 = Original quantity demanded

Q_1 = New quantity demanded

P_0 = Original price

P_1 = New price

We can take a numerical example to illustrate arc elasticity. Suppose that the price of a commodity is Br. 5 and the quantity demanded at that price is 100 units of a commodity. Now assume that the price of the commodity falls to Br. 4 and the quantity demanded rises to 110 units. In terms of the above formula, the value of the arc elasticity will be

$$E_d = -3/7$$

Note that:

- ✓ Elasticity of demand is unit free because it is a ratio of percentage change.
- ✓ Elasticity of demand is usually a negative number because of the law of demand. If the price elasticity of demand is positive the product is inferior.
- If $E_d \geq |1|$, demand is said to be elastic and the product is luxury product
- If $E_d \leq |1|$, demand is inelastic and the product is necessity
- If $E_d = |1|$, demand is unitary elastic.
- If $E_d = |0|$, demand is said to be perfectly inelastic.
- If $E_d = |\infty|$ demand is said to be perfectly elastic.

Determinants of price Elasticity of Demand

The following factors make price elasticity of demand elastic or inelastic other than changes in the price of the product.

- i) **The availability of substitutes:** the more substitutes available for a product, the more *elastic* will be the price elasticity of demand.
- ii) **Time:** In the long- run, price elasticity of demand tends to be elastic. Because:
 - More substitute goods could be produced.
 - People tend to adjust their consumption pattern.
- iii) **The proportion of income consumers spend for a product:-**the smaller the proportion of income spent for a good, the less price elastic will be.
- iv) **The importance of the commodity in the consumers' budget :**
 - Luxury goods tend to be more **elastic**, example: gold.
 - Necessity goods tend to be **less elastic** example: Salt.

B. Income Elasticity of Demand

It is a measure of responsiveness of demand to change in income.

$$Id = \frac{\% \Delta Q}{\% \Delta I} = \frac{(Q_1 - Q_0)}{I_1 - I_0} \frac{(I_0)}{(Q_0)}$$

Point income elasticity of demand:

- i) If $Id \geq 1$, the good is **luxury** good.
- ii) If $Id \leq 1$ but positive, the good is **necessity** good,
- iii) If Id is negative, the good is **inferior** good.

C. Cross price Elasticity of Demand

Measures how much the demand for a product is affected by a change in price of another good.

$$E_{XY} = \frac{\% \Delta Q_X}{\% \Delta P_Y} = \frac{\Delta Q_X}{\Delta P_Y} \frac{(P_{0Y})}{(Q_{0X})} .$$

- i) The cross – price elasticity of demand for substitute goods is **positive**.
- ii) The cross – price elasticity of demand for complementary goods is **negative**. iii)
The cross – price elasticity of demand for unrelated goods is **zero**.

Example: Consider the following data which shows the changes in quantity demanded of good X in response to changes in the price of good Y.

Unit price of Y	Quantity demanded of X
10	1500
15	1000

Calculate the cross –price elasticity of demand between the two goods. What can you say about the two goods?

Check that the two goods are complements.

2.2 Theory of supply

Supply indicates various quantities of a product that sellers (producers) are willing and able to provide at different prices in a given period of time, other things remaining unchanged.

The law of supply: states that, ceteris paribus, as price of a product increase, quantity supplied of the product increases, and as price decreases, quantity supplied decreases. It tells us there is a positive relationship between price and quantity supplied.

2.2.1 Supply schedule, supply curve and supply function

A supply schedule is a tabular statement that states the different quantities of a commodity offered for sale at different prices.

Table 2.3: an individual seller's supply schedule for butter

Price (birr per kg)	30	25	20	15	10
Quantity supplied kg/week	100	90	80	70	60

A supply curve conveys the same information as a supply schedule. But it shows the information graphically rather than in a tabular form.

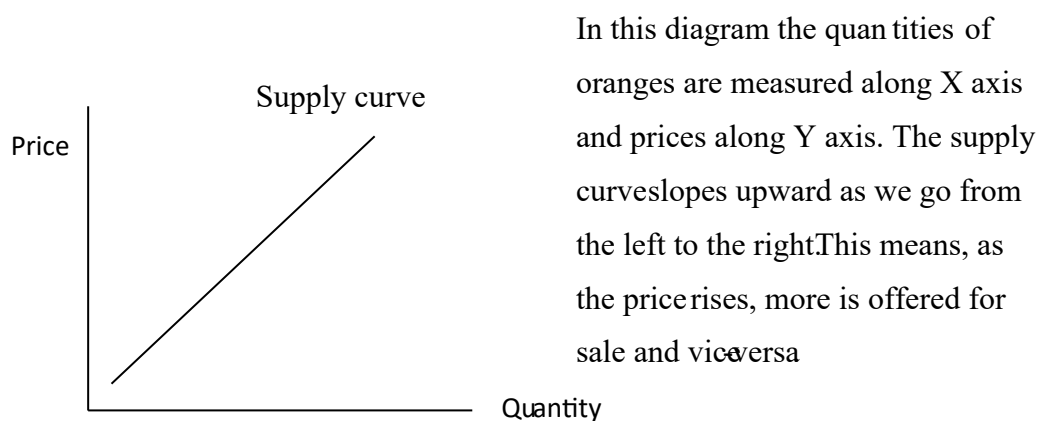


Figure 2.5 supply curve

The supply of a commodity can be briefly expressed in the following functional relationship:

$S = f(P)$, where S is quantity supplied and P is price of the commodity.

Market supply: It is derived by horizontally adding the quantity supplied of the product by all sellers at each price.

Table 2.4: Derivation of the market supply of good X

Price per unit	Quantity supplied by seller 1	Quantity supplied by seller 2	Quantity supplied by seller 3	Market supply per week
5	11	15	8	34
4	10.5	13	7	30.5
3	8	11.5	5.5	25
2	6	8.5	4	18.5
1	4	6	2	12

2.2.2 Determinants of supply

Apart from the change in price which causes a change in quantity demanded, the supply of a particular product is determined by:

- i) price of inputs (cost of inputs)
- ii) technology
- iii) prices of related goods iv) sellers' expectation of price of the product
- v) taxes & subsidies vi) number of sellers in the market
- vii) weather, etc.

i) Effect of change in input price on supply of a product

An increase in the price of inputs such as labour, raw materials, capital, etc causes a decrease in the supply of the product which is represented by a leftward shift of the supply curve. Likewise, a decrease in input price causes an increase in supply.

ii) Effect of change in Technology

Technological advancement enables a firm to produce and supply more in the market. This shifts the supply curve outward.

iii) Effect of change in weather condition

A change in weather condition will have an impact on the supply of a number of products, especially agricultural products. For example, other things remain unchanged, good weather condition boosts the supply of agricultural products. This shifts the supply curve of a given agricultural product outward. Bad weather condition will have the opposite impact.

Activity: Discuss how supply is affected by the changes in prices of related goods, taxes & subsidies, sellers' expectations of future price of the product, and the number of sellers in the market?

2.2.3 Elasticity of supply

It is the degree of responsiveness of the supply to change in price. It may be defined as the percentage change in quantity supplied divided by the percentage change in price. As the case with price elasticity of demand, we can measure the price elasticity of supply using point and arc elasticity methods. However, a simple and most commonly used method is point method.

The point price elasticity of supply can be calculated as the ratio of proportionate change in quantity supplied of a commodity to a given proportionate change in its price. Thus, the formula for measuring price elasticity of supply is:

$$E_s = \frac{\% \text{ change in quantity supplied}}{\% \text{ change in price}} = \left(\frac{Q_2 - Q_1}{p_2 - p_1} \right) \left(\frac{P_0}{Q_0} \right)$$

Like elasticity of demand, price elasticity of supply can be elastic, inelastic, unitary elastic, perfectly elastic or perfectly inelastic. The supply is elastic when a small change on price leads to great change in supply. It is inelastic or less elastic when a great change in price induces only a slight change in supply. If the supply is *perfectly inelastic*, it will be represented by a vertical line shown as below. If supply is *perfectly elastic* it will be represented by a horizontal straight line as in second diagram.

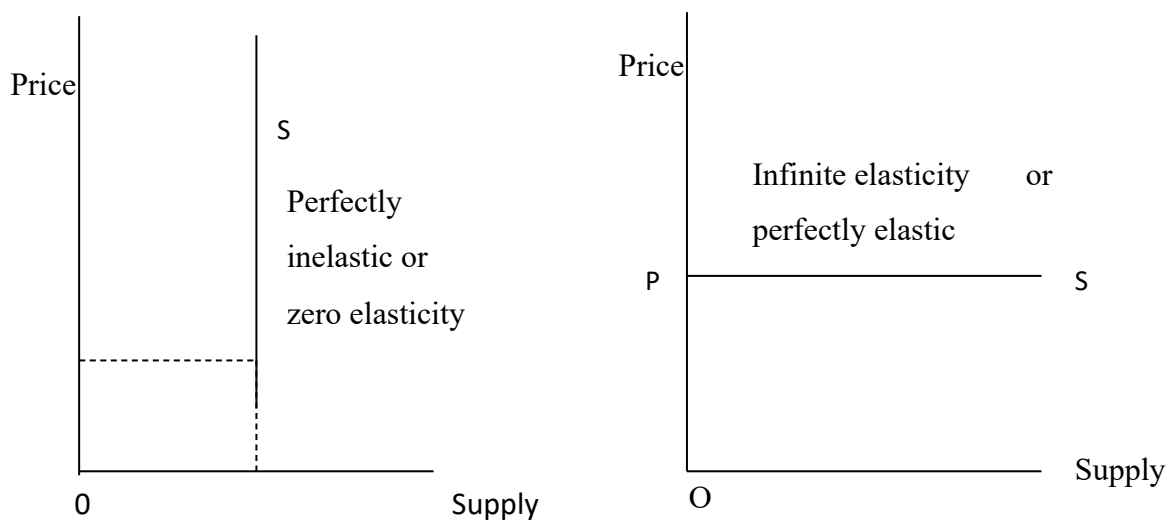


Figure 2.6: Perfectly inelastic and perfectly elastic supply curves

2.3 Market equilibrium

Having seen the demand and supply side of the market, now let's bring demand and supply together so as to see how the market price of a product is determined. Market equilibrium occurs when market demand equals market supply.

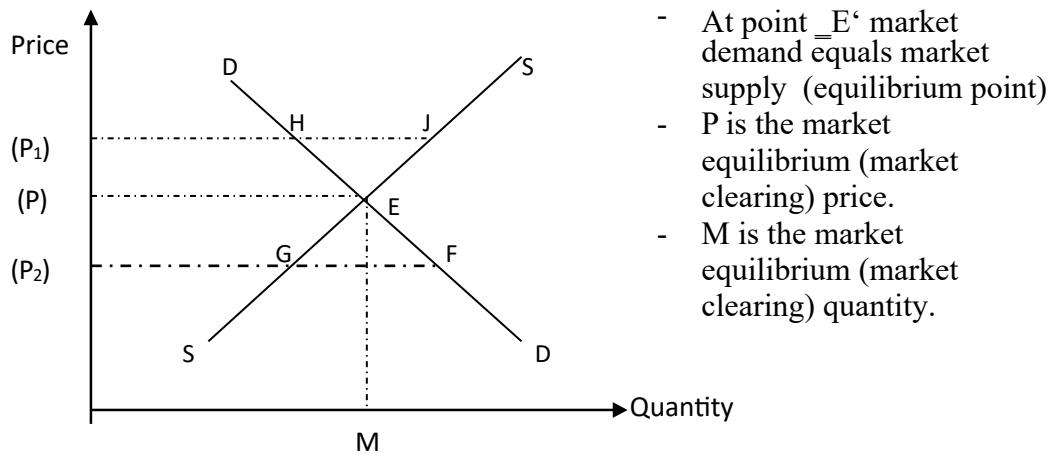


Figure 2.7: market equilibrium

In the above graph, any price greater than P will lead to market surplus. As the price of the commodity increases, consumers demand less of the product. On the other hand, as the price of increases, producers supply more of the good. Therefore, if price increases to P_1 the market will have a surplus of HJ. If the price decreases to P_2 buyers demand to buy more and suppliers prefer to decrease their supply leading to shortage in the market which is equal to GF.

Numerical example: Given market demand: $Q_d = 100 - 2P$, and market supply: $P = (Q_s / 2) + 10$

- Calculate the market equilibrium price and quantity
- Determine, whether there is surplus or shortage at $P = 25$ and $P = 35$.

Solution:

- At equilibrium, $Q_d = Q_s$

$$100 - 2P = 2P - 20$$

$$4P = 120$$

$$\underline{\underline{P = 30}}, \text{ and } \underline{\underline{Q = 40}}$$

- $Q_d(\text{at } P = 25) = 100 - 2(25) = 50$ and $Q_s(\text{at } P = 25) = 2(25) - 20 = 30$

Therefore, there is a shortage of: $50 - 30 = \underline{\underline{20 \text{ units}}}$

$Q_d(\text{at } P = 35) = 100 - 2(35) = 30$ and $Q_s(\text{at } p = 35) = 2(35) - 20 = 50$, a surplus of **20 units**

Effects of shift in demand and supply on equilibrium

Given demand and supply the equilibrium price and quantity are stable. However, when these market forces change what will happen to the equilibrium price and quantity? Changes in demand and supply bring about changes in the equilibrium price level and the equilibrium quantity.

i) when demand changes and supply remains constant

Factors such as changes in income, tastes, and prices of related goods will lead to a change in demand. The figure below shows the effects of a change in demand and the resultant equilibrium price and quantity. DD is the demand curve and SS is the supply curve.

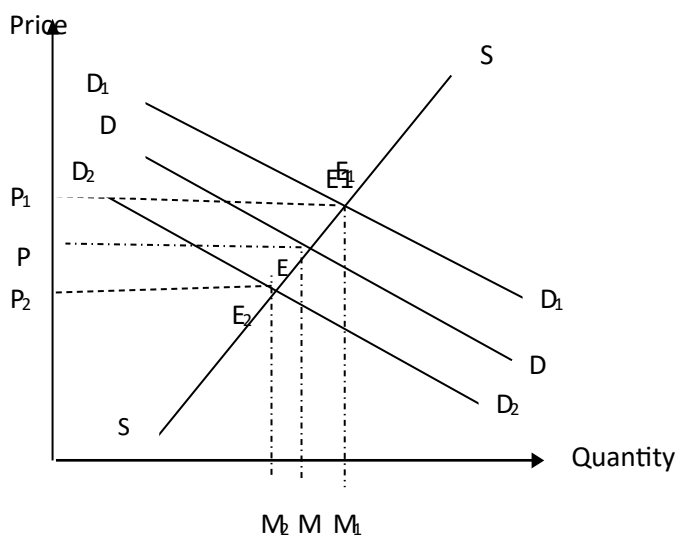


Figure 2.8: The effect of change in demand on market equilibrium

DD and SS curves intersect at point E and the quantity demanded and supplied is OM at OP equilibrium price. Given the supply, if the demand increases the demand curve will shift upward to the right. Due to a change in demand, the demand curve D_1D_1 intersects SS supply curve at point E_1 . The equilibrium price increases from OP to OP_1 and the equilibrium quantity from OM to OM_1 . On the other hand, if demand falls, the demand curve shifts downwards to the left. Due to a change in demand, the curve D_2D_2 intersects the supply curve SS at point E_2 . The equilibrium price decreases from OP to OP_2 and the equilibrium quantity decreases from OM to OM_2 . Supply being given, a decrease in demand reduces both the equilibrium price and the quantity and vice versa.

ii. When supply changes and demand remains constant

Changes in supply are brought by changes in technical knowledge and factor prices. The following graph explains the effects of changes in supply.

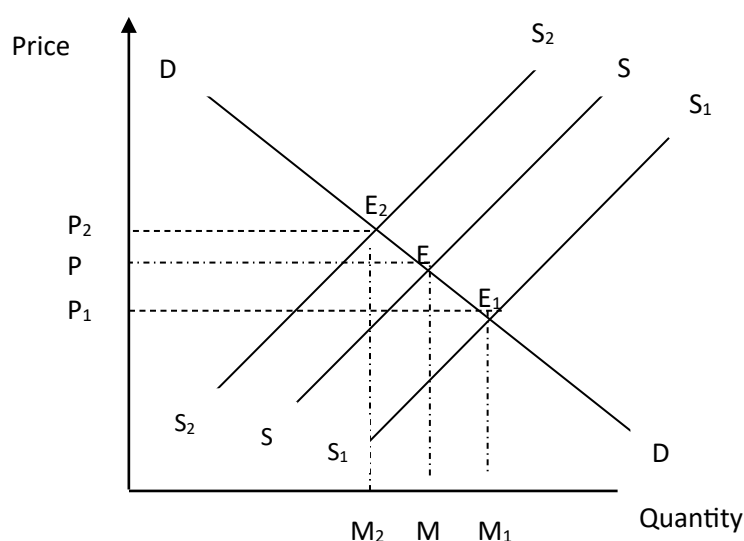


Figure 2.9: The effect of change in supply on market equilibrium

SS and DD intersect at point E, where supply and demand are equal at OM quantity at OP equilibrium price. Given the demand, if the supply increases, the supply curve shifts to the right (S_1S). The new supply curve, which intersects DD curve at E_1 , reduces the equilibrium price from OP to OP_1 and increases the equilibrium quantity from OM to OM_1 . On the contrary, when the supply falls, the supply curve moves to the left (S_2S) and intersects the DD curve at point E_2 raising the equilibrium price from OP to OP_2 and reducing the equilibrium quantity from OM to OM_2 .

III) Effects of combined changes in demand and supply

When both demand and supply increase, the quantity of the product will increase definitely. But it is not certain whether the price will rise or fall. If an increase in demand is more than an increase in supply, then the price goes up. On the other hand, if an increase in supply is more

than an increase in demand, the price falls but the quantity increases. If the increase in demand and supply is same, then the price remains the same.

When demand and supply decline, the quantity decreases. But the change in price will depend upon the relative fall in demand and supply. When the fall in demand is more than the fall in supply, the price will decrease. On the other hand, when the fall in supply is more than the fall in demand, the price will rise. If both demand and supply decline in the same ratio, there is no change in the equilibrium price, but the quantity decreases.

Activity: Considering the initial market equilibrium of figure 2.9 above, show the new equilibrium

1. if there is an increase in supply and proportionate increase in demand
2. if the magnitude of an increment in demand is less than an increment in supply
3. if demand and supply change in the opposite directions

Chapter summary

Demand for a commodity refers to the amount that will be purchased at a particular price during a particular period of time. Price of the commodity, income of the consumer, prices of related goods, consumer's tastes and preferences, consumers' expectations and number of buyers are considered the main determinants of demand for a commodity. *The law of demand* states that, other things remaining constant, the quantity demanded of a commodity increases when its price falls and decreases when the price rises.

Supply refers to the quantity of a commodity which producers are willing to produce and offer for sale at a particular price during a particular period of time. Price of a commodity, input prices, prices of related products, techniques of production, policy of taxation and subsidy, expectations of future prices, and the number of sellers are the main determinants of supply. Law of supply states that other things remaining the same, the quantity of any commodity that firms will produce and offer for sale rises with a rise in price and falls with a fall in price.

Market equilibrium refers to a situation in which quantity demanded of a commodity equals the quantity supplied of a commodity.

Goods can be categorized as *normal good* (a good for which the demand increases with increases in income), *an inferior good* (a good for which the demand tends to fall with an increase in the income of the consumer), *substitute goods* (are those goods which satisfy the same type of demand and can be used in place of one another), *complementary goods* (are those goods which are used jointly or together), and *giffen goods* (whose demand falls with a fall in their prices).

Elasticity of demand refers to the degree of responsiveness of quantity demanded of a commodity to change in any of its determinants. There are three types of elasticity of demand: *Price elasticity of demand*, *income elasticity of demand* and *cross price elasticity of demand*. Price elasticity of demand is determined by *availability of substitutes*, *nature of the commodity*, *proportion of income spent* and *time*.

Chapter Three

Theory of Consumer Behaviour

Introduction

In our day –to– day life, we buy different goods and services for consumption. As consumer, we act to derive satisfaction by using goods and services. But, have ever thought of how your mother or any other person whom you know decides to buy those consumption goods and services? Consumer theory is based on what people like, so it begins with something that we can't directly measure, but must infer. That is, consumer theory is based on the premise that we can infer what people like from the choices they make.

Consumer behaviour can be best understood in three steps. First, by examining consumer's preference, we need a practical way to describe how people prefer one good to another. Second, we must take into account that consumers face budget constraints – they have limited

incomes that restrict the quantities of goods they can buy. Third, we will put consumer preference and budget constraint together to determine consumer choice.

Chapter objectives

After successful completion of this chapter, you will be able to:

- explain consumer preferences and utility
- differentiate between cardinal and ordinal utility approach
- define indifference curve and discuss its properties
- derive and explain the budget line
- describe the equilibrium condition of a consumer

3.1 Consumer preferences

A consumer makes choices by comparing bundle of goods. Given any two consumption bundles, the consumer either decides that one of the consumption bundles is strictly better than the other, or decides that she is indifferent between the two bundles.

In order to tell whether one bundle is preferred to another, we see how the consumer behaves in choice situations involving two bundles. If she always chooses X when Y is available, then it is natural to say that this consumer prefers X to Y. We use the symbol $\succ$ to mean that one bundle is strictly preferred to another, so that $X \succ Y$ should be interpreted as saying that the consumer strictly prefers X to Y, in the sense that she definitely wants the X-bundle rather than the Y-bundle. If the consumer is indifferent between two bundles of goods, we use the symbol $\sim$ and write $X \sim Y$. Indifference means that the consumer would be just as satisfied, according to her own preferences, consuming the bundle X as she would be consuming bundle Y. If the consumer prefers or is indifferent between the two bundles we say that she weakly prefers X to Y and write $X \succeq Y$.

The relations of strict preference, weak preference, and indifference are not independent concepts; the relations are themselves related. For example, if $X \succeq Y$ and $Y \succeq X$, we can

conclude that $X \sim Y$. That is, if the consumer thinks that X is at least as good as Y and that Y is at least as good as X , then she must be indifferent between the two bundles of goods. Similarly, if $X \geq Y$ but we know that it is not the case that $X \sim Y$, we can conclude that $X > Y$. This just says that if the consumer thinks that X is at least as good as Y , and she is not indifferent between the two bundles, then she thinks that X is strictly better than Y .

3.2 The concept of utility

Economists use the term *utility* to describe the satisfaction or pleasure derived from the consumption of a good or service. In other words, utility is the power of the product to satisfy human wants. Given any two consumption bundles X and Y , the consumer definitely wants the X -bundle than the Y -bundle if and only if the utility of X is better than the utility of Y .

Do you think that utility and usefulness are synonymous? Do two individuals always derive equal satisfaction from consuming the same level of a product?

In defining utility, it is important to bear in mind the following points.

- *‘Utility’ and ‘Usefulness’ are not synonymous.* For example, paintings by Picasso may be useless functionally but offer great utility to art lovers. Hence, *usefulness* is product centric whereas *utility* is consumer centric.
- *Utility is subjective.* The utility of a product will vary from person to person. That means, the utility that two individuals derive from consuming the same level of a product may not be the same. For example, non-smokers do not derive any utility from cigarettes.
- *Utility can be different at different places and time.* For example, the utility that we get from drinking coffee early in the morning may be different from the utility we get during lunch time.

3.3 Approaches of measuring utility

How do you measure or compare the level of satisfaction (utility) that you obtain from goods and services?

There are two major approaches to measure or compare consumer's utility: cardinal and ordinal approaches. The cardinalist school postulated that utility can be measured objectively. According to the ordinalist school, utility is not measurable in cardinal numbers rather the consumer can rank or order the utility he derives from different goods and services.

3.3.1 The cardinal utility theory

According to the cardinal utility theory, utility is measurable by arbitrary unit of measurement called **utils** in the form of 1, 2, 3 etc. For example, we may say that consumption of an orange gives Bilen 10 utils and a banana gives her 8 utils, and so on. From this, we can assert that Bilen gets more satisfaction from orange than from banana.

3.3.1.1 Assumptions of cardinal utility theory

The cardinal approach is based on the following major assumptions.

1. **Rationality of consumers.** The main objective of the consumer is to maximize his/her satisfaction given his/her limited budget or income. Thus, in order to maximize his/her satisfaction, the consumer has to be rational.
2. **Utility is cardinally measurable.** According to the cardinal approach, the utility or satisfaction of each commodity is measurable. Utility is measured in subjective units called utils.

3. **Constant marginal utility of money.** A given unit of money deserves the same value at any time or place it is to be spent. A person at the start of the month where he has received monthly salary gives equal value to 1 birr with what he may give it after three weeks or so.
4. **Diminishing marginal utility (DMU).** The utility derived from each successive units of a commodity diminishes. In other words, the marginal utility of a commodity diminishes as the consumer acquires larger quantities of it.
5. **The total utility of a basket of goods depends on the quantities of the individual commodities.** If there are n commodities in the bundle with quantities $X_1, X_2, \dots, X_n$, the total utility is given by $TU = f(X_1, X_2, \dots, X_n)$.

3.3.1.2 Total and marginal utility

Total Utility (TU) is the total satisfaction a consumer gets from consuming some specific quantities of a commodity at a particular time. As the consumer consumes more of a good per time period, his/her total utility increases. However, there is a saturation point for that commodity beyond which the consumer will not be capable of enjoying any greater satisfaction from it.

Marginal Utility (MU) is the *extra satisfaction* a consumer realizes from an additional unit of the product. In other words, marginal utility is the change in total utility that results from the consumption of one more unit of a product. Graphically, it is the slope of total utility.

Mathematically, marginal utility is:

$$MU = \frac{\Delta TU}{\Delta Q}$$

where, ΔTU is the change in total utility, and ΔQ is the change in the amount of product consumed.

To explain the relationship between TU and MU, let us consider the following hypothetical example.

Table 3.1: Total and marginal utility

Quantity	Total utility (TU)	Marginal utility (MU)
0	0	-
1	10	10
2	18	8
3	24	6
4	28	4
5	30	2
6	30	0
7	28	-2

The total utility first increases, reaches the maximum (when the consumer consumes 6 units) and then declines as the quantity consumed increases. On the other hand, the marginal utility continuously declines (even becomes zero or negative) as quantity consumed increases.

Graphically, the above data can be depicted as follows.

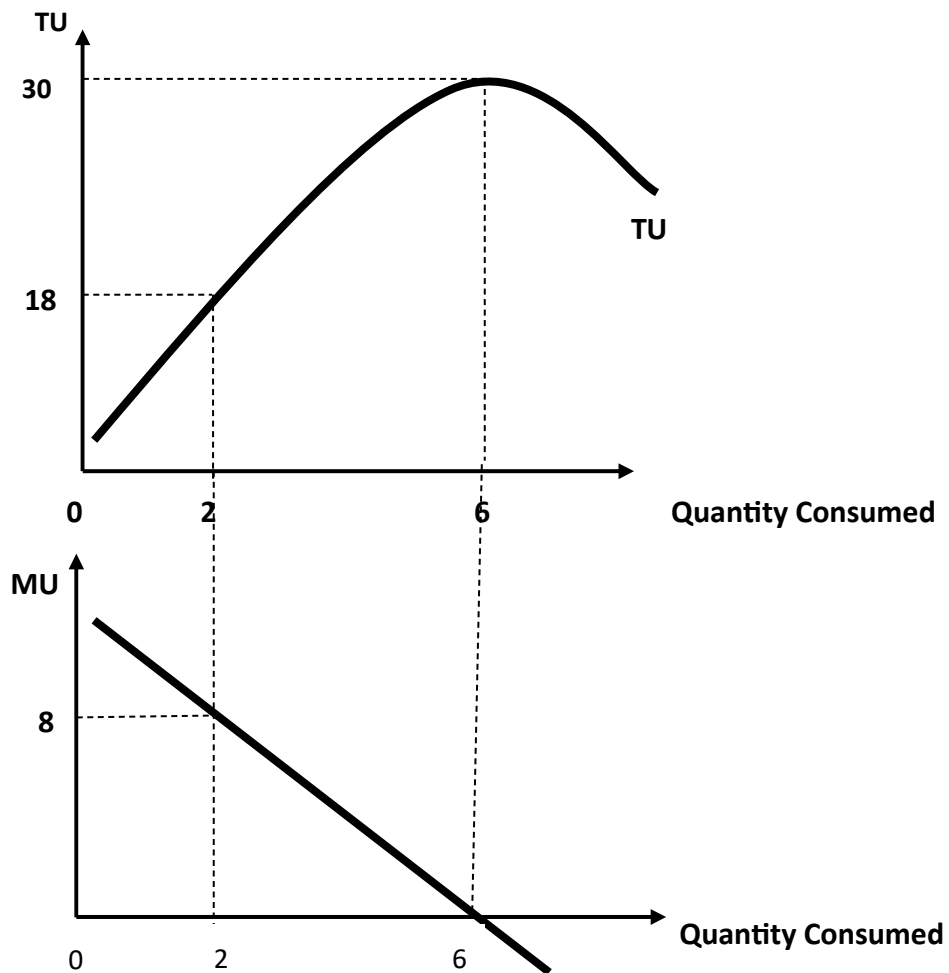


Figure 3.1: Total and marginal utility curves

As it can be observed from the above figure,

- When TU is increasing, MU is positive.
- When TU is maximized, MU is zero.
- When TU is decreasing, MU is negative.

3.3.1.3 Law of diminishing marginal utility (LDMU)

Is the utility you get from consumption of the first orange the same as the second or the third orange?

The *law of diminishing marginal utility* states that as the quantity consumed of a commodity increases per unit of time, the utility derived from each successive unit decreases, consumption of all other commodities remaining constant. In other words, the extra satisfaction that a consumer derives declines as he/she consumes more and more of the product in a given period of time. This gives sense in that the first banana a person consumes gives him more marginal utility than the second and the second banana also gives him higher marginal utility than the third and so on (see figure 3.1).

The law of diminishing marginal utility is based on the following assumptions.

- The consumer is rational
- The consumer consumes identical or homogenous product. The commodity to be consumed should have similar quality, color, design, etc.
- There is no time gap in consumption of the good
- The consumer taste/preferences remain unchanged

3.3.1.4 Equilibrium of a consumer

The objective of a rational consumer is to maximize total utility. As long as the additional unit consumed brings a positive marginal utility, the consumer wants to consume more of the product because total utility increases. However, given his limited income and the price level of goods and services, what combination of goods and services should he consume so as to get the maximum total utility?

a) the case of one commodity

The equilibrium condition of a consumer that consumes a single good X occurs when the marginal utility of X is equal to its market price.

$$MU_X = P_X$$

Proof

Given the utility function

$$U = f(X)$$

If the consumer buys commodity X, then his expenditure will be $Q_X P_X$. The consumer maximizes the difference between his utility and expenditure. $Max(U - Q_X P_X)$

The necessary condition for maximization is equating the derivative of a function to zero.

Thus, $\frac{dTU}{dQ_X} = 0$

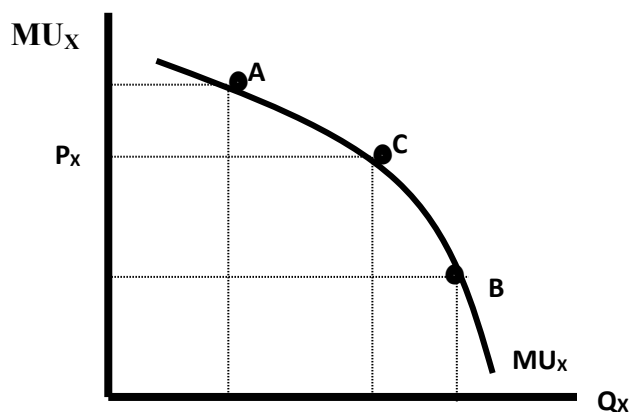


Figure 3.2: Equilibrium condition of consumer with only one commodity

At any point above point C (like point A) where $MU_X > P_X$, it pays the consumer to consume more. When $MU_X < P_X$ (like point B), the consumer should consume less of X. At point C where $MU_X = P_X$ the consumer is at equilibrium.

b) The case of two or more commodities

For the case of two or more goods, the consumer's equilibrium is achieved when the marginal utility per money spent is equal for each good purchased and his money income available for the purchase of the goods is exhausted. That is,

$$\frac{MU_X}{P_X} = \frac{MU_Y}{P_Y} = \dots = \frac{MU_N}{P_N} \text{ and } P_X Q_X + P_Y Q_Y + \dots + P_N Q_N = M$$

where, M is the income of the consumer.

Example: Suppose Saron has 7 Birr to be spent on two goods: banana and bread. The unit price of banana is 1 Birr and the unit price of a loaf of bread is 4 Birr. The total utility she obtains from consumption of each good is given below.

Table 3.2: Utility schedule for two commodities

<i>Income = 7 Birr, Price of banana = 1 Birr, Price of bread = 4 Birr</i>							
<i>Banana</i>				<i>Bread</i>			
<i>Quantity</i>	<i>TU</i>	<i>MU</i>	<i>MU/P</i>	<i>Quantity</i>	<i>TU</i>	<i>MU</i>	<i>MU/P</i>
0	0	-	-	0	0	-	-
1	6	6	6	1	12	12	3
2	11	5	5	2	20	8	2
3	14	3	3	3	26	6	1.5
4	16	2	2	4	29	3	0.75
5	16	0	0	5	31	2	0.5
6	14	-2	-2	6	32	1	0.25

Recall that utility is maximized when the condition of marginal utility of one commodity divided by its market price is equal to the marginal utility of the other commodity divided by its market price.

$$\frac{MU_1}{P_1} = \frac{MU_2}{P_2}$$

In table 3.2, there are two different combinations of the two goods where the MU of the last birr spent on each commodity is equal. However, only one of the two combinations is consistent with the prices of the goods and her income. Saron will be at equilibrium when she consumes 3 units of banana and 1 loaf of bread. At this equilibrium,

i) $MU_1/P_1 = MU_2/P_2$

$$\frac{MU_{\text{banana}}}{P_{\text{banana}}} = \frac{MU_{\text{bread}}}{P_{\text{bread}}} \quad \text{---}$$

ii) $P_1.Q_1 + P_2.Q_2 = M$

$$(1 \times 3) + (4 \times 1) = 7$$

The total utility that Saron derives from this combination can be given by:

$$TU = TU_1 + TU_2$$

$$TU = 14 + 12$$

$$TU = 26$$

Given her fixed income and the price level of the two goods, no combination of the two goods will give her higher TU than this level of utility.

Limitation of the cardinal approach

1. The assumption of cardinal utility is doubtful because utility may not be quantified. Utility cannot be measured absolutely (objectively).
2. The assumption of constant MU of money is unrealistic because as income increases, the marginal utility of money changes.

3.3.2 The ordinal utility theory

In the ordinal utility approach, it is not possible for consumers to express the utility of various commodities they consume in absolute terms, like 1 util, 2 utils, or 3 utils but it is possible to express the utility in relative terms. The consumers can rank commodities in the order of their preferences as 1st, 2nd, 3rd and so on. Therefore, the consumer need not know in specific units the utility of various commodities to make his choice. It suffices for him to be able to rank the various baskets of goods according to the satisfaction that each bundle gives him.

3.3.2.1 Assumptions of ordinal utility theory

The ordinal approach is based on the following assumptions.

- **Consumers are rational** - they maximize their satisfaction or utility given their income and market prices.
- **Utility is ordinal** - utility is not absolutely (cardinally) measurable. Consumers are required only to order or rank their preference for various bundles of commodities.

- **Diminishing marginal rate of substitution:** The marginal rate of substitution is the rate at which a consumer is willing to substitute one commodity for another commodity so that his total satisfaction remains the same. The rate at which one good can be substituted for another in consumer's basket of goods diminishes as the consumer consumes more and more of the good.
- The **total utility** of a consumer is measured by the amount (quantities) of all items he/she consumes from his/her **consumption basket**.
- **Consumer's preferences are consistent.** For example, if there are three goods in a given consumer's basket, say, X, Y, Z and if he prefers X to Y and Y to Z, then the consumer is expected to prefer X to Z. This property is known as axioms of **transitivity**.

The ordinal utility approach is explained with the help of indifference curves. Therefore, the ordinal utility theory is also known as the *indifference curve approach*.

3.3.2.2 Indifference set, curve and map

Indifference set/ schedule is a combination of goods for which the consumer is indifferent. It shows the various combinations of goods from which the consumer derives the same level of satisfaction.

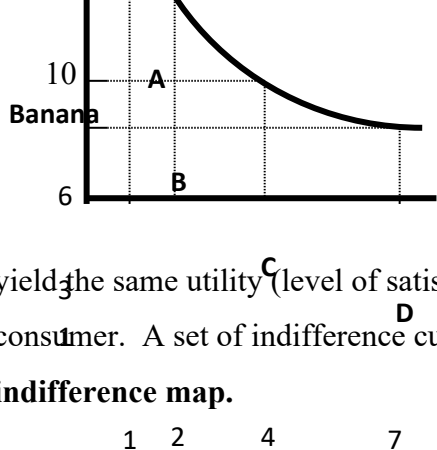
Consider a consumer who consumes two goods X and Y (table 3.3).

Table 3.3: Indifference schedule

Bundle (Combination)	A	B	C	D
Orange	1	2	4	7
Banana	10	6	3	1

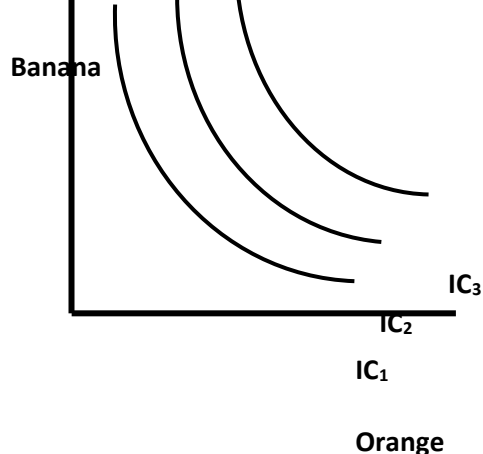
In table 3.3 above, each combination of good X and Y gives the consumer equal level of total utility. Thus, the individual is indifferent whether he consumes combination A, B, C or D.

Indifference curve: When the indifference set/schedule is expressed graphically, it is called an *indifference curve*. An indifference curve shows different combinations of two goods which



yield the same utility (level of satisfaction) to the consumer. A set of indifference curves is called

indifference map.



i) Indifference curve

ii) Indifference map

Figure 3.3: Indifference curve and indifference map

3.3.2.3 Properties of indifference curves

- 1. Indifference curves have negative slope (downward sloping to the right).** Indifference curves are negatively sloped because the consumption level of one commodity can be increased only by reducing the consumption level of the other commodity. In other words, in order to keep the utility of the consumer constant, as the quantity of one commodity is increased the quantity of the other must be decreased.
- 2. Indifference curves are convex to the origin.** This implies that the slope of an indifference curve decreases (in absolute terms) as we move along the curve from the left downwards to the right. The convexity of indifference curves is the reflection of the diminishing marginal rate of substitution. This assumption implies that the commodities can substitute one another at any point on an indifference curve but are not perfect substitutes.
- 3. A higher indifference curve is always preferred to a lower one.** The further away from the origin an indifferent curve lies, the higher the level of utility it denotes. Baskets of goods on a higher indifference curve are preferred by the rational consumer because they contain more of the two commodities than the lower ones.
- 4. Indifference curves never cross each other (cannot intersect).** The assumptions of consistency and transitivity will rule out the intersection of indifference curves. Figure 3.4 shows the violations of the assumptions of preferences due to the intersection of indifference curves.

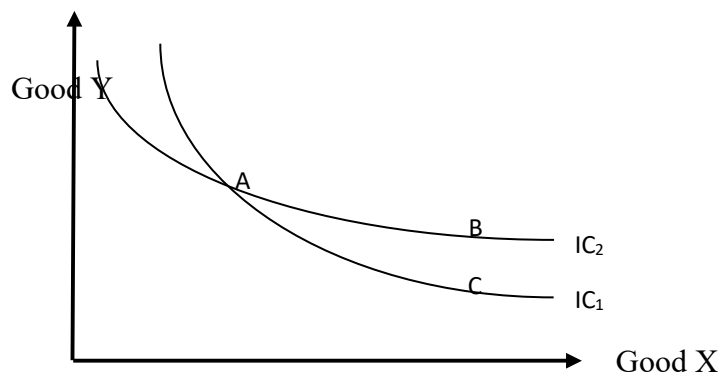


Figure 3.4: Intersection of indifference curves

In the above figure, the consumer prefers bundle B to bundle C. On the other hand, following indifference curve 1 (IC1), the consumer is indifferent between bundle A and C, and along indifference curve 2 (IC2) the consumer is indifferent between bundle A and B. According to the principle of transitivity, this implies that the consumer is indifferent between bundle B and C which is contradictory or inconsistent with the initial statement where the consumer prefers bundle B to C. Therefore, indifference curves never cross each other.

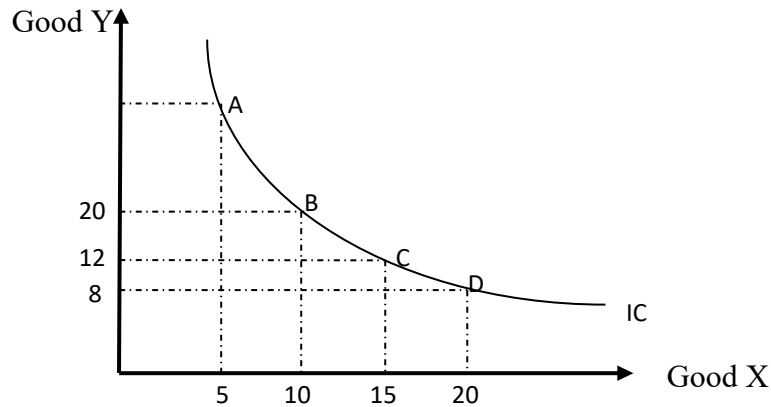
22 3.3.2.4 Marginal rate of substitution (MRS)

Marginal rate of substitution is a rate at which consumers are willing to substitute one commodity for another in such a way that the consumer remains on the same indifference curve. It shows a consumer's willingness to substitute one good for another while he/she is indifferent between the bundles.

Marginal rate of substitution of X for Y is defined as the number of units of commodity Y that must be given up in exchange for an extra unit of commodity X so that the consumer maintains the same level of satisfaction. Since one of the goods is sacrificed to obtain more of the other good, the MRS is negative. Hence, usually we take the absolute value of the slope.

$$MRS_{X,Y} = \frac{\text{Number of units of Y given up}}{\text{Number of units of X gained}} = \frac{\Delta Y}{\Delta X}$$

To understand the concept, consider the following indifference curve.



30

Figure 3.5: Indifference curve for two products X and Y

From the above graph, $MRS_{X,Y}$ associated with the movement from point A to B, point B to C and point C to D is 2.0, 1.6, and 0.8 respectively. That is, for the same increase in the consumption of good X, the amount of good Y the consumer is willing to sacrifice diminishes. This principle of marginal rate of substitution is reflected by the convex shape of the indifference curve and is called diminishing marginal rate of substitution.

It is also possible to derive MRS using the concept of marginal utility. $MRS_{X,Y}$ is related to MU_X and MU_Y as follows.

$$MRS_{X,Y} = \frac{MU_X}{MU_Y}$$

Example: Suppose a consumer's utility function is given by $U(X,Y) = X^4 Y^2$. Find $MRS_{X,Y}$

—

3.3.2.5 The budget line or the price line

Do you think that the indifference curve discussed in the previous section tells us whether a given combination of goods is affordable to the consumer? If no, what are the major constraints to the consumer in maximizing his/her total utility?

Indifference curves only tell us about consumer preferences for any two goods but they cannot show which combinations of the two goods will be bought. In reality, the consumer is constrained by his/her income and prices of the two commodities. This constraint is often presented with the help of the budget line.

The *budget line* is a set of the commodity bundles that can be purchased if the entire income is spent. It is a graph which shows the various combinations of two goods that a consumer can purchase given his/her limited income and the prices of the two goods.

In order to draw a budget line facing a consumer, we consider the following assumptions.

- There are only two goods bought in quantities, say, X and Y .
- Each consumer is confronted with market determined prices, P_X and P_Y .
- The consumer has a known and fixed money income (M).

Assuming that the consumer spends all his/her income on the two goods (X and Y), we can express the budget constraint as:

$$M = P_X X + P_Y Y$$

By rearranging the above equation, we can derive the following general equation of a budget line.

$$Y = M/P_Y - P_X/P_Y X$$

Graphically,

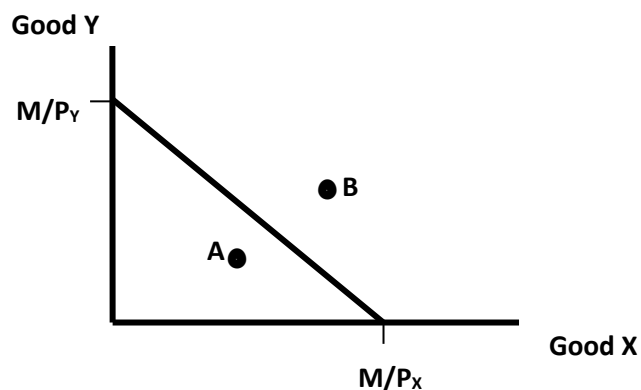


Figure 3.6: The budget line

Note that: The slope of the budget line is given by $-P_X/P_Y$ (the ratio of the prices of the two goods).

- Any combination of the two goods within the budget line (such as point A) or along the budget line is attainable.
- Any combination of the two goods outside the budget line (such as point B) is unattainable (unaffordable).

Example: A consumer has \$100 to spend on two goods X and Y with prices \$3 and \$5 respectively. Derive the equation of the budget line and sketch the graph.

Solution: The equation of the budget line can be derived as follows.

$$P_X X + P_Y Y = M$$

$$3X + 5Y = 100$$

$$5Y = 100 - 3X$$



$$Y = 20 - 3/5X$$

When the consumer spends all of her income on good Y, we get the Y- intercept (0,20).

Similarly, when the consumer spends all of her income on good X, we obtain the X- intercept (33.3,0). Using these two points we can sketch the graph of the budget line.

Recall that a budget is drawn for given prices and fixed consumer's income. Hence, the changes in prices or income will affect the budget line.

Change in income: If the income of the consumer changes (keeping the prices of the commodities unchanged), the budget line also shifts (changes). Increase in income causes an upward/outward shift in the budget line that allows the consumer to buy more goods and services and decreases in income causes a downward/inward shift in the budget line that leads the consumer to buy less quantity of the two goods. It is important to note that the slope of the budget line (the ratio of the two prices) does not change when income rises or falls.

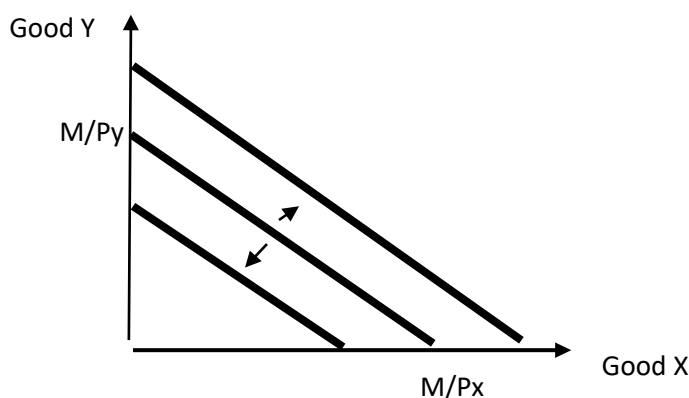


Figure 3.7: *Effects of increase (right) and decrease (left) in income on the budget line*

Change in prices: An equal increase in the prices of the two goods shifts the budget line inward. Since the two goods become expensive, the consumer can purchase the lesser amount of the two goods. An equal decrease in the prices of the two goods, on the other hand, shifts the

budget line out ward. Since the two goods become cheaper, the consumer can purchase the more amounts of the two goods.

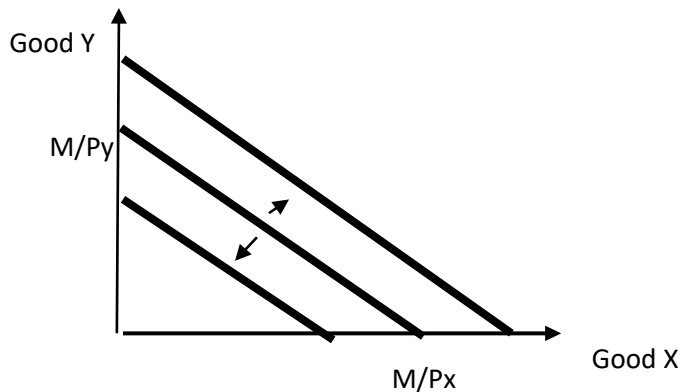


Figure 3.8: *Effect of proportionate increase (inward) and decrease (out ward) in the prices of both goods*

An increase or decrease in the price of one of the two goods, keeping the price of the other good and income constant, changes the slope of the budget line by affecting only the intercept of the commodity that records the change in the price. For instance, if the price of good X decreases while both the price of good Y and consumer's income remain unchanged, the horizontal intercept moves outward and makes the budget line flatter. The reverse is true if the price of good X increases. On the other hand, if the price of good Y decreases while both the price of good X and consumer's income remain unchanged, the vertical intercept moves upward and makes the budget line steeper. The reverse is true for an increase in the price of good Y.

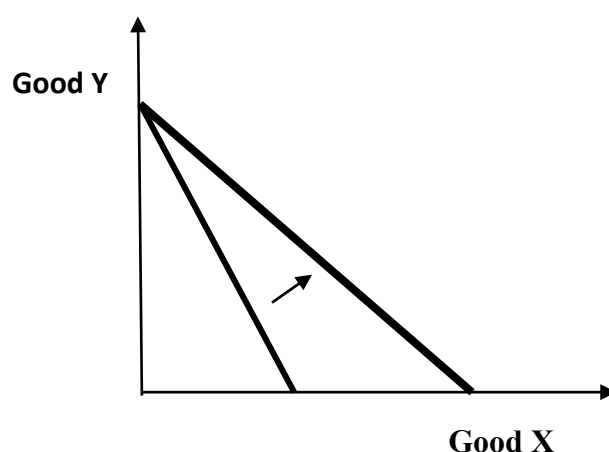


Figure 3.9: *Effect of decrease in the price of only good X on the budget line*

3.3.2.6 Equilibrium of the consumer

The preferences of a consumer (what he/she wishes to purchase) are indicated by the indifference curve. The budget line specifies different combinations of two goods (say X and Y) the consumer can purchase with the limited income. Therefore, a rational consumer tries to attain the highest possible indifference curve, given the budget line. This occurs at the point where the indifference curve is tangent to the budget line so that the slope of the indifference curve (MRS_{XY}) is equal to the slope of the budget line (P_X/P_Y). In figure 3.10, the equilibrium of the consumer is at point 'E' where the budget line is tangent to the highest attainable indifference curve (IC_2).

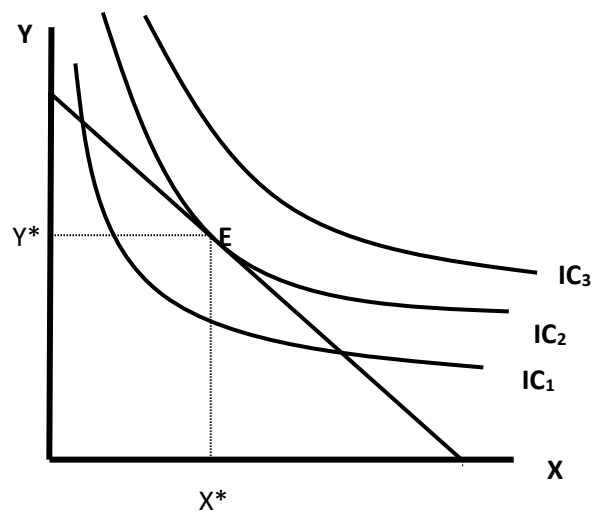


Figure 3.10: Consumer equilibrium under indifference curve approach

Mathematically, consumer optimum (equilibrium) is attained at the point where:

Slope of indifference curve = Slope of the budget line

$$MRS_{XY} = \frac{P_X}{P_Y}$$

Example: A consumer consuming two commodities X and Y has the utility function $U(X,Y) = XY + 2X$. The prices of the two commodities are 4 birr and 2 birr respectively. The consumer has a total income of 60 birr to be spent on the two goods.

- a) Find the utility maximizing quantities of good X and Y.
- b) Find the $MRS_{X,Y}$ at equilibrium.

Solution

- a) The budget constraint of the consumer is given by:

$$P_X \cdot X + P_Y \cdot Y = M$$

$$4X + 2Y = 60 \dots\dots\dots (i)$$

Moreover, at equilibrium

$$MUX / P_X = MUY / P_Y$$

Boom/peak: it is a phase in which the economy is producing the highest level of output in the business cycle. It is the point which marks the end of economic expansion and the beginning of recession. In this phase, the economy's output is growing faster than its long-term (potential) trend and is therefore unsustainable. Due to very high degree of utilization of resources, unemployment level is low; business is good; and it is a period of prosperity.